

# Atom- og Molekylefysik Formelsamling

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## 1 Sandwichformlen

In quantum mechanics, dipole transitions involve matrix elements of the form:

$$\langle \psi_f | r_i | \psi_i \rangle, \quad (1)$$

where  $r_i = x, y$ , or  $z$  represents the position operator components in spherical coordinates.

The goal is to determine which operator ( $x, y$ , or  $z$ ) to use based on selection rules and symmetry.

Step 1: Check the Selection Rules Dipole transitions must satisfy the following:

1. **Change in orbital angular momentum:**

$$\Delta l = \pm 1. \quad (2)$$

2. **Change in magnetic quantum number  $m$ :**

$$\Delta m = 0, \pm 1. \quad (3)$$

3. Different operators couple different values of  $m$ :

- $x$  and  $y$  couple  $m \rightarrow m \pm 1$ .
- $z$  couples  $m \rightarrow m$  (i.e.,  $\Delta m = 0$ ).

Step 2: Transformation of  $x, y, z$  in Terms of Spherical Harmonics. In spherical coordinates, the position operators transform as:

$$x = r \sin \theta \cos \phi \quad (\text{behaves like } Y_1^{+1} + Y_1^{-1}) \quad (4)$$

$$y = r \sin \theta \sin \phi \quad (\text{behaves like } i(Y_1^{+1} - Y_1^{-1})) \quad (5)$$

$$z = r \cos \theta \quad (\text{behaves like } Y_1^0). \quad (6)$$

This means:

- $z$  can only couple states with  $\Delta m = 0$ .
- $x$  and  $y$  couple states with  $\Delta m = \pm 1$ .
- $y$  introduces an imaginary factor  $i$ , while  $x$  does not.

Step 3: Match  $\Delta m$  to the Right Operator. The choice of operator depends on the magnetic quantum number change:

Step 4: Check for Integral Symmetry. Even if the selection rules allow a transition, the integral may still vanish due to symmetry:

- If the integral contains an **odd function over a symmetric range**, it is **zero**.
- If the integral contains an **orthogonal spherical harmonic combination**, it is **zero**.
- If the integral using  $x$  is zero, try  $y$  instead. The extra **imaginary factor** in  $y$  can sometimes fix cancellations.

| Initial $m$ | Final $m$ | Use Operator |
|-------------|-----------|--------------|
| 0           | 0         | $z$          |
| 0           | $\pm 1$   | $x$ or $y$   |
| $\pm 1$     | 0         | $x$ or $y$   |
| $\pm 1$     | $\pm 1$   | Forbidden    |
| $\pm 1$     | $\mp 1$   | $x$ or $y$   |
| Any         | $\pm 2$   | Forbidden    |

Tabel 1: Operator Selection Based on  $\Delta m$

Example Applications Example 1:  $\langle \psi_{100} | x | \psi_{211} \rangle$

- $\psi_{100}$  has  $m = 0$ .
- $\psi_{211}$  has  $m = +1$ .
- $\Delta m = +1 \Rightarrow$  Use  $x$  or  $y$ .
- The  $x$ -integral vanishes, but  $y$  introduces an  $i$  factor and gives a nonzero result.

**Final choice:** Use  $y$  instead of  $x$ .

Example 2:  $\langle \psi_{100} | z | \psi_{210} \rangle$

- $\psi_{100}$  has  $m = 0$ .
- $\psi_{210}$  has  $m = 0$ .
- $\Delta m = 0 \Rightarrow$  Use  $z$ .
- The integral is nonzero.

**Final choice:** Use  $z$ .

Example 3:  $\langle \psi_{100} | y | \psi_{21-1} \rangle$

- $\psi_{100}$  has  $m = 0$ .
- $\psi_{21-1}$  has  $m = -1$ .
- $\Delta m = -1 \Rightarrow$  Use  $x$  or  $y$ .
- The  $x$ -integral vanishes, but  $y$  introduces an imaginary phase.

**Final choice:** Use  $y$ .

Summary

1. **Check selection rules:** Ensure  $\Delta l = \pm 1$  and  $\Delta m = 0, \pm 1$ .
2. **Match  $\Delta m$  to the correct operator:**
  - $\Delta m = 0 \Rightarrow$  Use  $z$ .
  - $\Delta m = \pm 1 \Rightarrow$  Use  $x$  or  $y$ .
3. **Check spherical harmonics for integral symmetry:** If  $x$ -integral is zero, try  $y$ .
4. **Use the decision table to quickly determine the valid operator.**